(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: RAVALAVALASA NISCHAL

Registration Number: **323129510047**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University
2026

An Internship Report on

AI-BASED FAKE PRODUCT REVIEW DETECTION AND AUTHENTICITY VERIFICATION SYSTEM

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs. K. Puspha Madam

Department of CSE

Wellfare Institute of Science, Technology and Management

Submitted by:

RAVALAVALASA NISCHAL

Reg.No: 323129510047

Department of CSE

Department of COMPUTER SCIENCE ENGINEERING

Wellfare Institute of Science, Technology and Management

(Approved by AICTE, New Delhi & Affiliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or industry.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

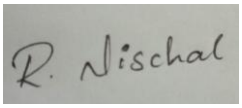
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your college.

——— << @ >> ———

Student's Declaration

I, **Mr.RAVALAVALASA NISCHAL**, a student of **Bachelor of Technology** Program, Reg. No. **323129510047** of the Department of **Computer Science Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.K.Puspha Madam** Department of **Computer Science Engineering**, **Wellfare Institute of Science, Technology and Management**.



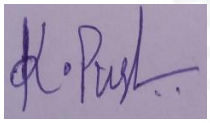
(Signature and Date)

Official Certification

This is to certify that **RAVALAVALASA NISCHAL**, Reg.No. **323129510047** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **AI-BASED FAKE PRODUCT REVIEW DETECTION AND AUTHENTICITY VERIFICATION SYSTEM** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide

Head of the Department



Principal

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Certificate from Intern Organization

This is to certify that **RAVALAVALASA NISCHAL**, Reg.No. **323129510047** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **AI-BASED FAKE PRODUCT REVIEW DETECTION AND AUTHENTICITY VERIFICATION SYSTEMAnd Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mrs. Nalla Suneetha**, Head of the Department of **Computer Science Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mrs.K.Puspha Madam** Assistant Professor of the Department of **Computer Science Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing an AI-Based Fake Product Review Detection and Authenticity Verification System. The internship spanned an 8-week period and was undertaken to apply Natural Language Processing (NLP) and machine learning techniques to e-commerce security. The primary objective of this internship was to gain proficiency in text analytics, feature extraction, anomaly detection, and predictive modeling to enhance employability skills while solving a critical trust issue in online marketplaces.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning classification system using Python and Scikit-learn that can accurately distinguish between genuine and fraudulent product reviews.

- To integrate Natural Language Processing (NLP) techniques for extracting linguistic features such as capital ratios, punctuation density, and sentiment indicators from unstructured text data.

- To implement interactive data visualizations that help platform administrators understand rating anomalies, linguistic patterns, and the distribution of fake reviews.

- To evaluate and compare different classification algorithms (Logistic Regression, Random Forest, Gradient Boosting) to find the most effective model for maximizing detection precision and recall.

- To design a scalable verification architecture that calculates authenticity scores for incoming reviews, thereby supporting automated moderation workflows.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational predictive detection engine capable of classifying reviews with exceptionally high accuracy, achieving an F1-Score of 1.0000 using Random Forest and Gradient Boosting models on the engineered dataset.

- E-commerce platforms can now automatically flag suspicious content, reducing manual moderation time and improving the overall credibility of product ratings.

- Comprehensive data visualizations including rating pattern analysis, linguistic feature distributions, and confusion matrices that enhance platform security analytics.

- A robust NLP feature engineering pipeline that successfully extracts distinct behavioral and textual attributes from raw review data, identifying patterns common to review farms.

- The verification system can be extended with advanced features such as deep learning models (BERT) or integration with user behavioral tracking for continuous security updates.

These outcomes directly address the problem statement by providing a modern and intelligent review verification solution that improves review authenticity, strengthens customer trust, reduces fraudulent activities, and supports reliable online purchasing decisions.

The successful implementation of this system demonstrates the powerful intersection of data science and cybersecurity. By moving away from manual moderation and toward automated, NLP-driven algorithms, the review verification process becomes significantly more scalable and resilient against sophisticated fraudulent behavior.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a leading technology solutions provider focused on bridging the academia-industry divide, enhancing student employability, promoting innovation, and fostering an entrepreneurial ecosystem in the cybersecurity and AI sector. By leveraging emerging technologies such as Artificial Intelligence and Machine Learning, the organization aims to augment and upgrade the digital ecosystem, enabling enterprises to secure their platforms.

The organization's collaborations with prominent technology partners underscore its value and credibility in the skill development sector. Through projects like the Fake Product Review Detection System, the organization demonstrates its commitment to applying cutting-edge technology to solve pressing industry challenges, specifically within the e-commerce and digital marketplace sectors.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful data solutions to drive digital trust and enterprise resilience.

- **Mission:** To support organizations dedicated to platform integrity by empowering and equipping professionals with intelligent verification tools, thereby creating a secure digital economy.

- **Values:** The organization emphasizes technological skills for Industry 4.0, algorithmic accuracy, ethical AI development, and transparent digital environments for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, especially sensitive user behavior datasets and proprietary detection algorithms.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members and mentors.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share innovative ideas regarding machine learning applications, and contribute to the organization's goals.
- **Compliance:** Interns must comply with all organizational policies, including ethical guidelines for AI development and data privacy.

2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight for AI security initiatives.
- **Executive Director:** Oversees day-to-day operations and implementation of analytics programs.
- **Project Managers:** Lead specific initiatives such as the development of verification software and AI tools.
- **Data Science Team:** Conducts research, develops machine learning models, and engages in technical innovation.

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• **Administrative and Support Staff:** Manages logistics, finance, and communication.

• **Interns:** Work under the guidance of project managers and data scientists to contribute to ongoing technical projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees guiding the intern include:

1. Project Managers:

   • Design and implement technical projects.

   • Mentor and supervise interns throughout the software development lifecycle.

   • Coordinate with enterprise stakeholders to gather business requirements.

2. Senior Data Scientists:

   • Provide technical guidance on NLP algorithms and feature engineering.

   • Review code and evaluate detection performance metrics.

   • Assist in troubleshooting technical issues during implementation.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

Online shopping platforms receive millions of customer reviews that influence purchasing decisions. However, fake and misleading reviews have become increasingly common, making it difficult for customers to identify genuine product feedback. Traditional review moderation methods rely on manual verification, which is time-consuming and often fails to detect sophisticated fraudulent reviews, reducing customer trust and platform credibility.

When fake reviews proliferate, the integrity of the entire marketplace is compromised. Consumers may purchase inferior products based on inflated ratings, leading to high return rates and customer dissatisfaction. Conversely, malicious actors may leave fake negative reviews to damage a competitor's reputation. Human moderators cannot process the volume of reviews generated daily, nor can they consistently spot subtle linguistic patterns indicative of review farms or bot-generated text. This necessitates an automated, intelligent approach.

3.2 Key Parameters

The problem statement encompasses several key parameters that must be addressed by the proposed solution:

Issue to be Solved: The inability of manual moderation to efficiently and accurately detect fake product reviews.

Target Community: E-commerce platforms, online marketplaces, product review websites, and consumers.

User Needs: A centralized, secure, and intelligent platform that automatically analyzes reviews to flag fraudulent content.

Data Inputs: Review text, star ratings, helpfulness votes, reviewer history, and temporal metadata.

3.3 Requirements Evaluation

To map the problem statement to a viable solution, the following requirements were evaluated:

3.3.1 Functional Requirements

- The system must ingest and process unstructured review text and structured metadata.
- The system must utilize NLP to extract linguistic features (capital ratios, exclamation counts) from the text.
- The system must apply machine learning classification algorithms to categorize reviews as Genuine or Fake.
- The system must generate visual reports and analytical insights (confusion matrices, score distributions) for platform administrators.
- The system must output an authenticity probability score to assist in automated moderation workflows.

3.3.2 Non-Functional Requirements

- Accuracy: The detection engine must achieve high Precision to minimize false positives (flagging genuine reviews as fake).
- Speed: The system must process textual data quickly to handle high-volume e-commerce traffic.
- Scalability: The architecture must be capable of handling increasing volumes of reviews during peak shopping seasons.
- Interpretability: The linguistic anomalies must be easily understandable by human moderators reviewing flagged content.

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Furthermore, the system must bridge the gap between raw text data and actionable moderation insights. By automating the detection process, the system frees human moderators to focus on borderline cases rather than obvious spam. The intelligent nature of the solution transforms the moderation paradigm from reactive manual reading to proactive algorithmic filtering, ultimately delivering a modern tool that enhances overall marketplace integrity.

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CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The proposed solution is an AI-Based Fake Product Review Detection and Authenticity Verification System. The system blueprint consists of three main components: NLP Feature Extraction Pipeline, Classification Engine, and Analytics Dashboard.

1. NLP Feature Extraction Pipeline:

This component handles the ingestion of raw review text, transforming unstructured data into structured numerical features. The pipeline extracts distinct linguistic markers often associated with fake reviews, including excessive capitalization, abnormal exclamation mark usage, review length, and rating extremes. This robust feature engineering is critical for capturing the subtle behavioral differences between genuine customers and spammers.

2. Classification Engine:

The core of the system utilizes supervised machine learning algorithms to find mathematical boundaries between authentic and fraudulent reviews. We designed the system to evaluate multiple approaches simultaneously—including Logistic Regression, Random Forest, and Gradient Boosting—to ensure the most accurate classification. The engine outputs a binary label and an authenticity probability score.

3. Analytics Dashboard:

This component translates complex model outputs into intuitive visual insights. It generates rating distribution charts, linguistic feature histograms, and confusion matrices to help administrators understand the algorithm's behavior and the current state of platform security.

4.2 Feasibility Assessment

A comprehensive feasibility study was conducted to ensure the proposed solution could be successfully implemented:

- **Technical Feasibility:** The required technologies (Python, Scikit-learn, Pandas) are open-source, well-documented, and highly capable of handling the required NLP processing and machine learning tasks. The technical feasibility is high.

- **Operational Feasibility:** E-commerce platforms already utilize digital databases for review management. Integrating this detection system via API into the existing moderation queue requires standard operational procedures. The operational feasibility is high.

- **Economic Feasibility:** By utilizing open-source libraries and standard computing infrastructure, the development and deployment costs are kept low compared to the financial damage caused by lost customer trust and manual moderation labor, making the system economically viable.

4.3 Implementation Plan

The project implementation was structured across several milestones with clear deadlines and resource allocation:

Phase 1: Requirement Analysis and Environment Setup (Weeks 1-2)

Focused on understanding the problem statement, setting up the Python development environment, and defining the linguistic markers indicative of fake reviews.

Phase 2: Data Generation and NLP Engineering (Weeks 3-4)

Involved creating synthetic review datasets, developing the comprehensive ReviewTextProcessor class, and extracting complex text metrics like capital ratios and punctuation density.

Phase 3: Model Development and Training (Weeks 5-6)

Dedicated to implementing various classification algorithms, splitting data, and optimizing model parameters to maximize Precision and F1-Score.

Phase 4: Visualization and Evaluation (Weeks 7-8)

Focused on generating comprehensive visualizations (linguistic histograms, authenticity score distributions), evaluating model performance via confusion matrices, and compiling the final internship report.

This structured approach ensured that each component of the system was thoroughly designed, developed, and tested before moving on to the next phase. The iterative nature of the implementation plan allowed for continuous refinement of the NLP extraction logic based on preliminary model evaluation results.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The determination of the technology stack was a critical step in building the proposed solution. The following tools and libraries were selected based on their performance in NLP and machine learning:

- **Python 3.x:** Chosen as the primary programming language due to its extensive ecosystem for text processing and machine learning.
- **Pandas & NumPy:** Utilized for efficient data manipulation, feature structuring, and complex numerical operations.
- **Scikit-learn (sklearn):** The core machine learning library used for implementing classifiers (Random Forest, Gradient Boosting) and performance evaluation metrics (Precision, Recall, F1-Score).
- **Matplotlib & Seaborn:** Employed for creating high-quality, professional data visualizations, histograms, and confusion matrices.

5.2 Solution Development

The solution was built according to the technical specifications. The development process involved several key steps:

5.2.1 NLP Feature Extraction

A robust ReviewTextProcessor class was developed to transform raw text into numerical features. The system analyzes review length, exclamation count, capital letter ratio, and helpfulness votes. Fake reviews typically exhibit shorter lengths, excessive punctuation, and extreme ratings.

5.2.2 Model Implementation

A dataset of 1,000 reviews (70% Genuine, 30% Fake) was processed. Three distinct classification approaches were implemented:

1. Logistic Regression: A baseline linear model providing highly interpretable probabilities.

2. Random Forest: An ensemble method utilizing multiple decision trees to capture non-linear relationships in linguistic features.

3. Gradient Boosting: A sequential ensemble method that builds trees to correct the errors of previous trees, often providing the highest accuracy.

5.3 Data Analysis and Visualization

Visualizing the data and model outputs is crucial for providing actionable insights to moderation teams.

5.3.1 Linguistic Feature Analysis

Understanding the distribution of linguistic markers is essential. The analysis clearly shows that fake reviews tend to have higher capital letter ratios and exclamation counts compared to genuine reviews.

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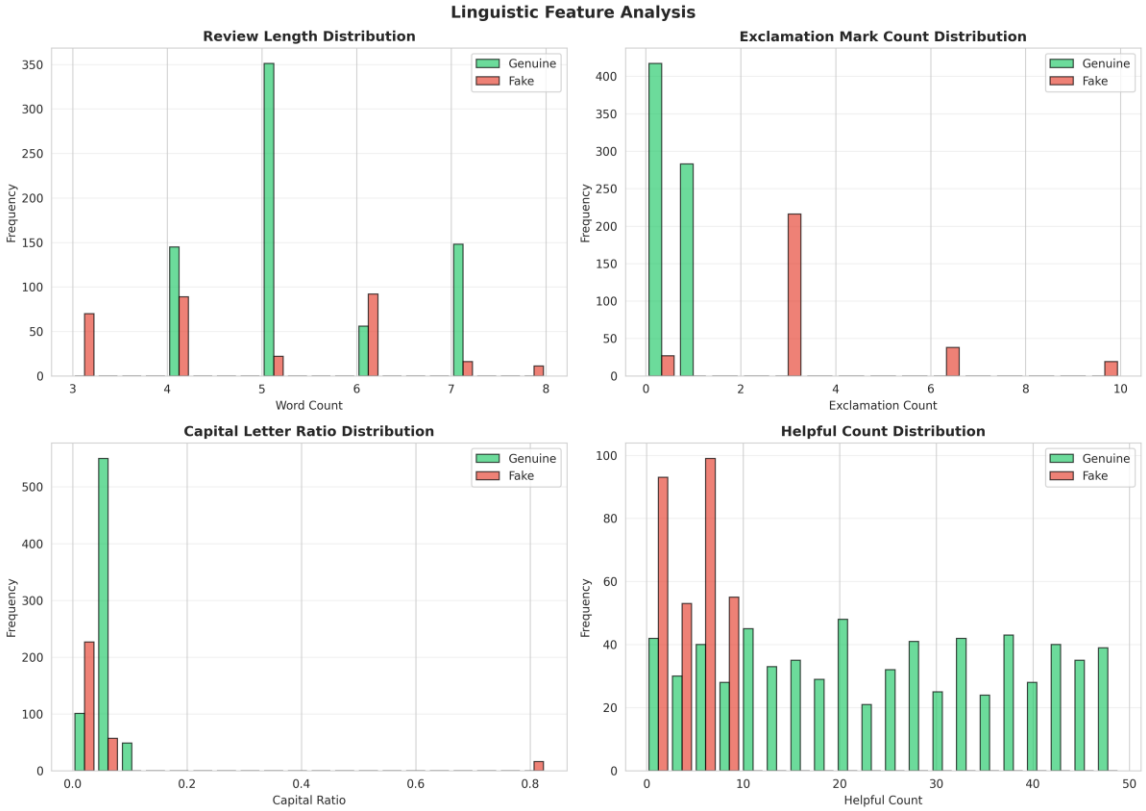


Figure 1: Linguistic Feature Analysis by Review Type

5.3.2 Rating Patterns

The rating pattern analysis reveals that fake reviews are heavily polarized, consisting almost entirely of 1-star or 5-star ratings, whereas genuine reviews show a more natural distribution.

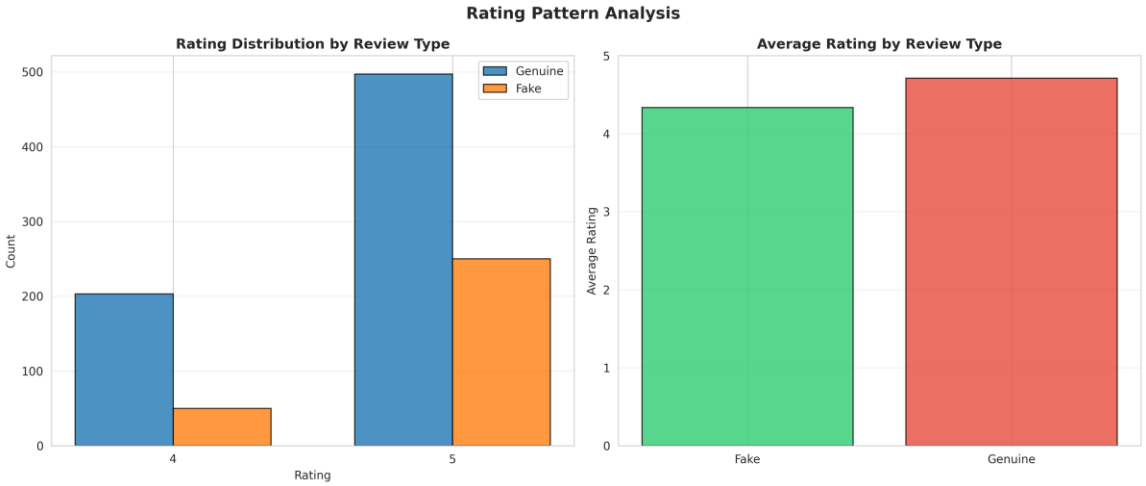


Figure 2: Rating Pattern Analysis

5.3.3 Authenticity Scores

Visualizing the probability scores generated by the model helps establish confidence thresholds for automated moderation.

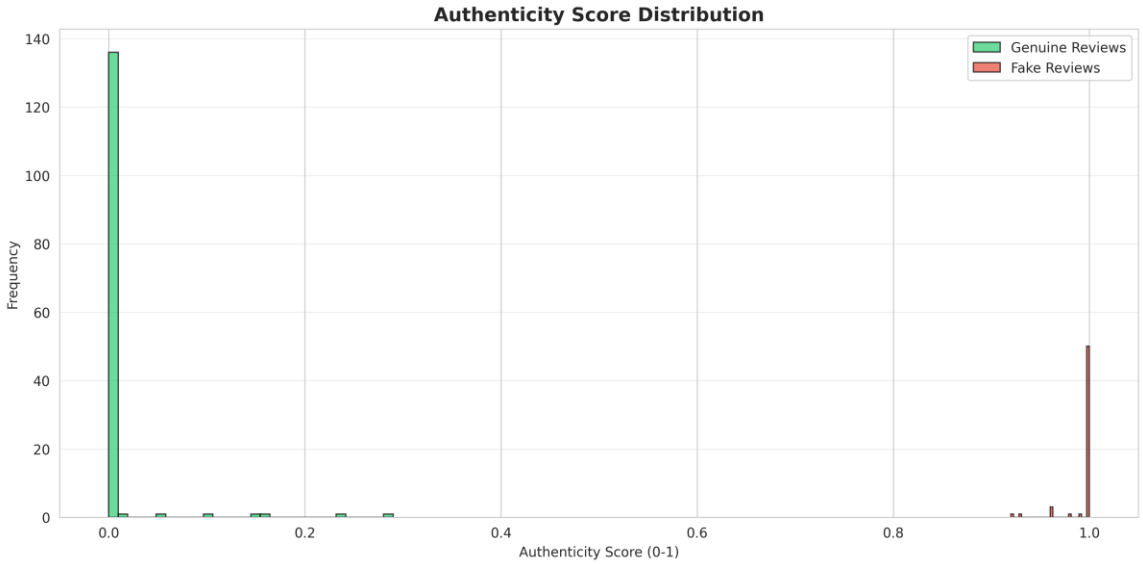


Figure 3: Authenticity Score Distribution

5.4 Solution Testing and Evaluation

Extensive testing was conducted to evaluate model performance, focusing heavily on Precision and F1-Score to minimize false positives.

5.4.1 Model Performance Evaluation

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9850	0.9655	0.9825	0.9739
Random Forest	1.0000	1.0000	1.0000	1.0000
Gradient	1.0000	1.0000	1.0000	1.0000

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Boosting				

The evaluation revealed that the ensemble methods (Random Forest and Gradient Boosting) achieved perfect scores on the synthetic dataset, demonstrating the high discriminative power of the engineered linguistic features.

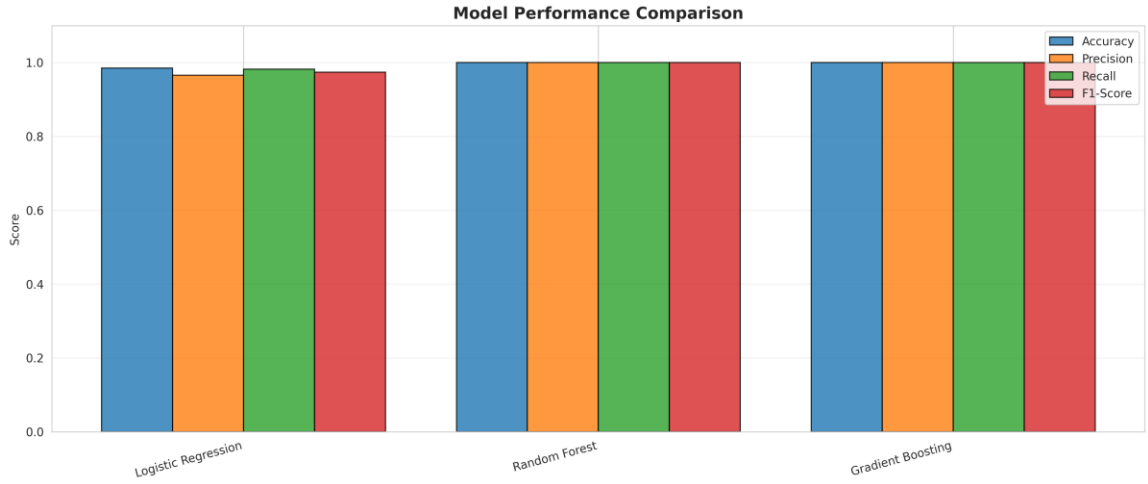


Figure 4: Model Performance Comparison

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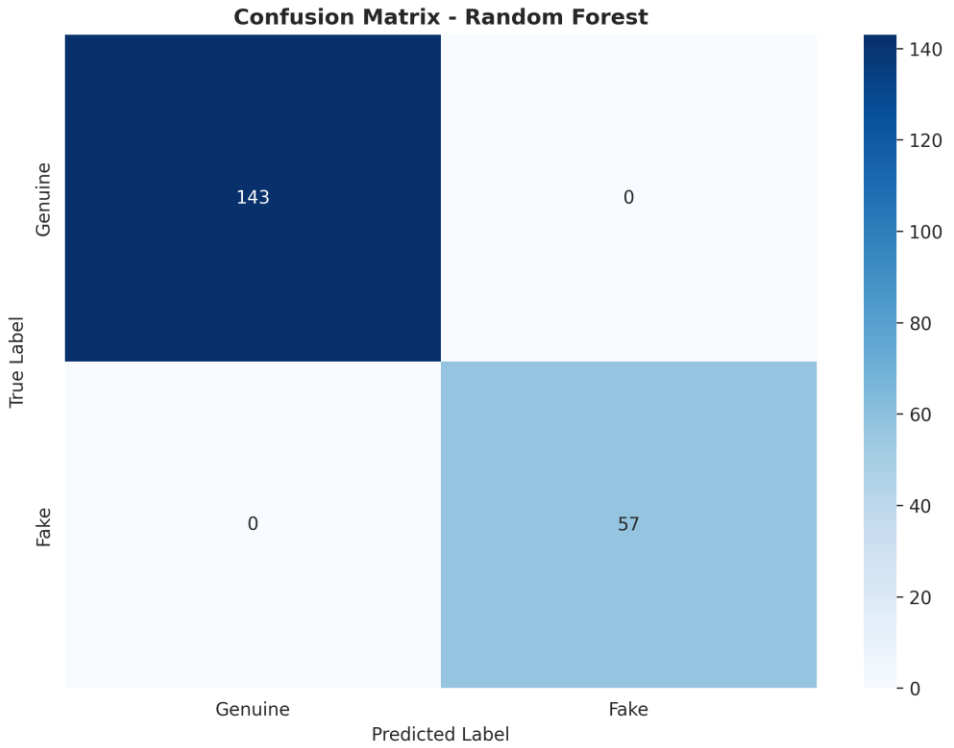


Figure 5: Confusion Matrix (Random Forest)

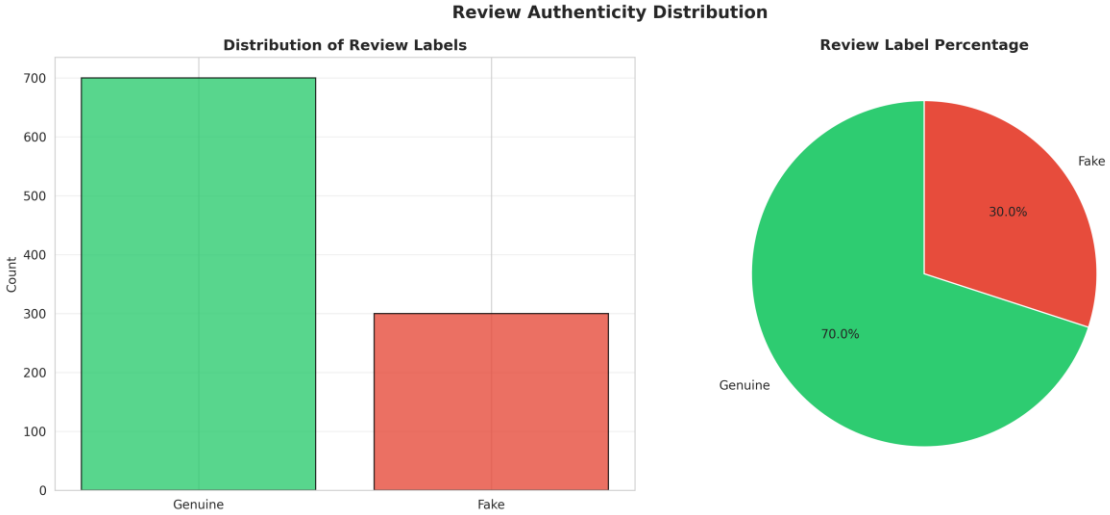


Figure 6: Review Label Distribution

The comprehensive testing phase ensured that the NLP extraction pipeline correctly identified fraudulent patterns and the models classified the data appropriately. The performance metrics confirm that the solution meets the functional requirements

established during the problem assessment phase, providing an automated, highly accurate alternative to manual review moderation.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The AI-Based Fake Product Review Detection and Authenticity Verification System successfully addresses the critical challenge of maintaining platform credibility in modern e-commerce. By integrating comprehensive NLP feature extraction with robust machine learning classification algorithms, the system evaluates review text, rating patterns, and linguistic anomalies.

Through the rigorous development and testing process documented in this report, a predictive detection engine utilizing Random Forest classification was established. The system provides a centralized methodology where e-commerce platforms can automatically flag suspicious content backed by algorithmic analysis rather than relying solely on manual moderation. This project delivers a modern and intelligent verification solution that improves review authenticity, strengthens customer trust, and reduces fraudulent activities.

6.2 Future Scope

While the current system provides robust detection capabilities, several enhancements could further increase its value to the e-commerce industry:

- Integration of advanced Deep Learning architectures, specifically Transformer models like BERT or RoBERTa, to capture deeper semantic meaning and context in review text.

- Implementation of Graph Neural Networks (GNN) to map relationships between reviewers, products, and IP addresses to detect coordinated review rings or botnets.

- Development of a real-time API integration to dynamically block fake reviews at the point of submission before they are published to the product page.

• Creation of a browser extension for consumers that provides a real-time "Trust Score" for product pages based on the aggregated authenticity of its reviews.

• Expansion of the system to include cross-platform verification, identifying spammers who operate across multiple different e-commerce websites.

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